

Lab 3: Component Modelling & Architectural Pattern

Selection Centralized Audit Trail Compliance Engine

Problem Statement #50 | Developer Tools & IT Operations

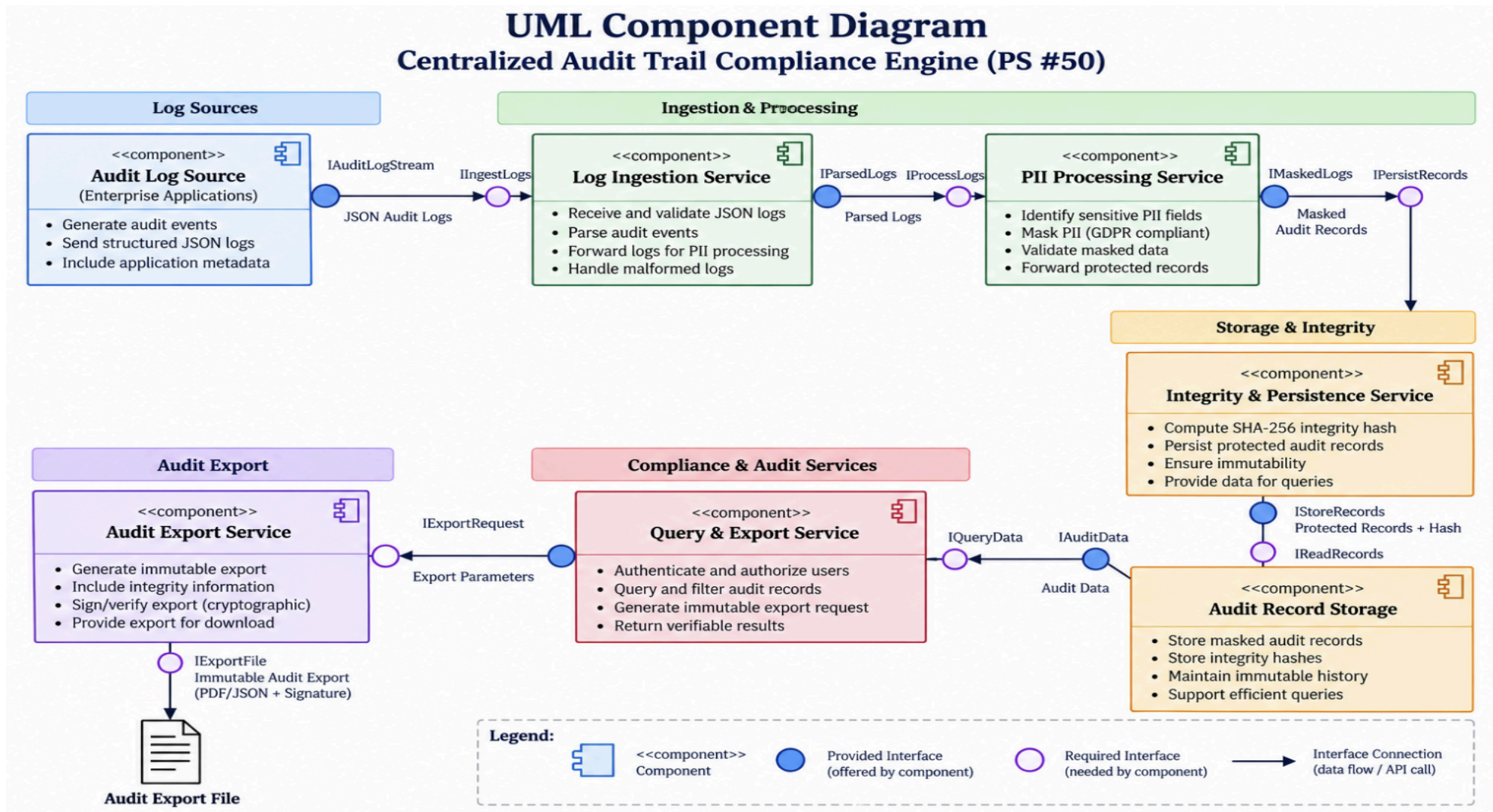


Figure 1. UML Component Diagram — Centralized Audit Trail Compliance Engine

Architecture Selection & Written Justification

Centralized Audit Trail Compliance Engine

Reason 1 — Clear Separation of Responsibilities

The system naturally separates into distinct responsibilities: **audit-log ingestion, JSON parsing and PII masking, protected record storage, integrity verification, and audit querying/export**. A layered structure keeps these responsibilities separated while maintaining a clear processing flow from incoming audit logs to protected storage and controlled retrieval.

Reason 2 — Maintainability and Clarity

Each major function has a focused responsibility and communicates through defined interfaces. This makes the architecture easier to understand, maintain, and modify when compliance or audit requirements evolve.

Reason 3 — Supports GDPR-Compliant PII Protection

The layered flow ensures that audit logs are **processed before persistence**. The JSON Parser & PII Masker identifies sensitive fields such as **Social Security Numbers and credit-card numbers** and masks them before the protected record reaches storage. This directly satisfies the requirement that PII must not be stored in cleartext.

Reason 4 — Strong Integrity and Tamper Detection

A dedicated **Integrity Verification Service** handles SHA-256 generation and verification. Persisted audit records therefore have associated integrity information, allowing tampering to be detected and supporting cryptographically verifiable audit exports.

Reason 5 — Efficient Historical Audit Queries

The **Audit Query & Export Service** is separated from ingestion and PII processing, so query and filtering operations can be optimized independently. This directly supports the requirement for complex filtering across **10 million historical records in under one second**.

Security Advantage

Layer separation ensures that **PII processing occurs before persistence**, reducing exposure of sensitive information. The separate integrity-verification responsibility also supports SHA-256 verification, tamper detection, and cryptographically verifiable exports.

Performance Benefit

Separating audit querying and filtering from ingestion and processing allows the query functionality to be optimized for large historical datasets. This is particularly important for the specified **10-million-record, sub-one-second query requirement**.

